

Multivariate Local Linear Regression

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Let $\mathbf{X}$ be a d -dimensional predictor taking values in $[0, 1]^d$, coupled with a response Y taking values in $\mathbb{R}$. Put $m(\cdot) := \mathbb{E}(Y \mid \mathbf{X} = \cdot)$ and denote by $\nabla_j(m)$ the partial derivative of m in the direction of the j -th coordinate:

$$\nabla_j(m)(\mathbf{x}) := \frac{\partial}{\partial x_j} m(\mathbf{x}).$$

For multivariate locally linear regression, we consider the following approximation of m around $\mathbf{x}$:

$$m(\mathbf{u}) \simeq m(\mathbf{x}) + \sum_{j=1}^d (u_j - x_j) \cdot \nabla_j(m)(\mathbf{x}), \quad \mathbf{u} \simeq \mathbf{x}.$$

Definition

We use a product kernel function, $\prod_{\ell=1}^d K_{\ell}(u_{\ell} - x_{\ell})$, for smoothing over $\mathbf{u}$ around $\mathbf{x}$. Multivariate local linear estimators of $(m, \nabla_1(m), \dots, \nabla_d(m))$:

$$\begin{pmatrix} \hat{m}(\mathbf{x}) \\ \hat{\nabla}_1(m)(\mathbf{x}) \\ \vdots \\ \hat{\nabla}_d(m)(\mathbf{x}) \end{pmatrix} := \arg \min_{a \in \mathbb{R}^{d+1}} n^{-1} \sum_{i=1}^n (y_i - a_0 - a_1(X_{i,1} - x_1) - \dots - a_d(X_{i,d} - x_d))^2 \prod_{\ell=1}^d K_{\ell}(X_{i,\ell} - x_{\ell}).$$

Again, we assume that $K \geq 0$ is bounded, supported on $[-1, 1]$, and satisfies $\int K = 1$ for simplicity.

Definition

Define the $n \times (d + 1)$ matrix D_x and $n \times n$ matrix W_x by

$$D_x^T = \begin{pmatrix} 1 & \cdots & 1 \\ \frac{X_{1,1}-x_1}{h_1} & \cdots & \frac{X_{n,1}-x_1}{h_1} \\ \vdots & \ddots & \vdots \\ \frac{X_{1,d}-x_d}{h_d} & \cdots & \frac{X_{n,d}-x_d}{h_d} \end{pmatrix}, \quad W_x = \text{diag} \left(\prod_{\ell=1}^d K_{\ell}(X_{i,\ell} - x_{\ell}) \right).$$

Then,

$$\begin{aligned} \hat{\mathbf{m}}(x) &= \begin{pmatrix} \hat{m}(\mathbf{x}) \\ h_1 \cdot \hat{\nabla}_1(m)(\mathbf{x}) \\ \vdots \\ h_d \cdot \hat{\nabla}_d(m)(\mathbf{x}) \end{pmatrix} := \arg \min_{a \in \mathbb{R}^{d+1}} (\mathbf{Y} - D_x a)^{\top} W_x (\mathbf{Y} - D_x a) \\ &= \left(D_x^{\top} W_x D_x \right)^{-1} D_x^{\top} W_x \mathbf{Y}. \end{aligned}$$

Multivariate version of Lemma 6.1

$\mathbf{X}_i \in \mathbb{R}^d$. For iid copies $(\mathbf{X}_i, \varepsilon_{ni})$, $1 \leq i \leq n$, with $\mathbb{E}(\varepsilon_{ni} \mid \mathbf{X}_i) = 0$. Assume that $\sup_n \mathbb{E}[\exp(\alpha|\varepsilon_{ni}|)] < \infty$ for some $\alpha > 0$, $\sup_n \mathbb{E}(\varepsilon_{ni}^2 \mid \mathbf{X}_i = \cdot)$ and p are bounded on $[0, 1]^d$. Suppose that K is Lipschitz continuous and

$$(nh_1 \dots h_d) / (\log n)^3 \rightarrow \infty \quad \text{as } n \rightarrow \infty.$$

Then, it holds that

$$\sup_{\mathbf{x} \in [0,1]^d} \left| n^{-1} \sum_{i=1}^n \left(\prod_{j=1}^d K_{h_j}(\mathbf{X}_{ij} - \mathbf{x}_j) \right) \varepsilon_{ni} \right| = O_p \left(n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \sqrt{\log n} \right).$$

simply, $K(\mathbf{u}; \mathbf{h}) = \prod_{\ell=1}^d K_{h_\ell}(u_\ell)$.

proof

Put $\tau_n = \sqrt{nh_1 \dots h_d / \log n}$. Define

$$Z_n(\mathbf{x}) = n^{-1} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) [\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) \mid \mathbf{X}_i)].$$

There exists a constant $0 < C < \infty$ such that

$$\mathbb{P} \left(\max_{1 \leq i \leq n} |\varepsilon_{ni}| \leq \tau_n \right) \geq 1 - C \cdot n \cdot \exp(-\alpha \tau_n) = 1 - C \cdot \exp \left[\log n (1 - \alpha \sqrt{\frac{nh_1 \dots h_d}{(\log n)^3}}) \right] \rightarrow 1.$$

This shows that

$$\sup_{\mathbf{x} \in [0,1]^d} \left| n^{-1} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) \right| = 0$$

with probability tending to one.

proof

Furthermore, we get that

$$\begin{aligned} & \sup_{\mathbf{x} \in [0,1]^d} \left| n^{-1} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \cdot \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) \mid \mathbf{X}_i) \right| \\ & \leq \frac{1}{h_1 \dots h_d} \cdot e^{-\alpha \tau_n / 2} \cdot (\sup K)^d \cdot \frac{2}{\alpha} \cdot n^{-1} \sum_{i=1}^n \mathbb{E}(e^{\alpha |\varepsilon_{ni}|} \mid X_i) \\ & \leq \frac{C}{\sqrt{nh_1 \dots h_d}} \exp \left(\frac{1}{2} \log \frac{n}{h_1 \dots h_d} - \frac{\alpha}{2} \sqrt{\frac{nh_1 \dots h_d}{\log n}} \right) \\ & \leq \frac{C}{\sqrt{nh_1 \dots h_d}} \exp \left(\log n - \frac{\alpha}{2} \sqrt{\frac{nh_1 \dots h_d}{\log n}} \right) = o\left(\frac{1}{\sqrt{nh_1 \dots h_d}}\right) \end{aligned}$$

with probability tending to one.

proof

We know that

$$\sup_{\mathbf{x} \in [0,1]^d} \left| n^{-1} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \cdot \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| > \tau_n) \mid \mathbf{X}_i) \right| = o_p(n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2}),$$

Using $\mathbb{E}(\varepsilon_{ni} \mid \mathbf{X}_i) = 0$,

$$Z_n(\mathbf{x}) = n^{-1} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) [\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) \mid \mathbf{X}_i)].$$

$$\sup_{\mathbf{x} \in [0,1]^d} \left| n^{-1} \sum_{i=1}^n \left(\prod_{j=1}^d K_{h_j}(\mathbf{X}_{ij} - \mathbf{x}_j) \right) \varepsilon_{ni} \right| = \sup_{x \in [0,1]} |Z_n(x)| + o_p(n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2}).$$

proof

Now, let $\mathcal{N}(\delta)$ be a δ -net such that, for any $x \in [0, 1]$, there exists a point $x' \in \mathcal{N}(\delta)$ such that $|x - x'| \leq \delta$. Claim

$$\sup_{\mathbf{x} \in \mathcal{N}(n^{-\gamma})^d} |Z_n(\mathbf{x})| = O_p(\tau_n^{-1})$$

for any $\gamma > 0$. This proves the lemma since

$$\begin{aligned} \sup_{\mathbf{x}, \mathbf{x}' \in [0,1]^d, |\mathbf{x}_i - \mathbf{x}'_i| \leq n^{-\gamma}} |Z_n(\mathbf{x}) - Z_n(\mathbf{x}')| &\leq C \cdot \frac{1}{h_1 \dots h_d} \cdot \frac{(n^{-\gamma})^d}{h_1 \dots h_d} \cdot n^{-1} \sum_{i=1}^n W_i \\ &\leq C' \frac{\tau_n}{h_1^2 \dots h_d^2 n^{\gamma d}} \end{aligned}$$

since $W_i = \varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) \mid \mathbf{X}_i)$ so that $|W_i| \leq 2\tau_n$.
and

$$\frac{\tau_n}{h_1^2 \dots h_d^2 n^{\gamma d}} \cdot \tau_n = n^{1-\gamma d} \frac{1}{h_1 \dots h_d \log n} \rightarrow 0 \text{ if } \gamma > 2/d$$

$$\sup_{\mathbf{x} \in \mathcal{N}(n^{-\gamma})^d} |Z_n(\mathbf{x})| = O_p(\tau_n^{-1})$$

follows if we prove

$$\sup_{\mathbf{x} \in [0,1]^d} \mathbb{P} \left(|Z_n(\mathbf{x})| > C \cdot \tau_n^{-1} \right) \leq C_0 \cdot n^{-g(C)}$$

for some constant $0 < C_0 < \infty$ and some function g such that $g(C) \uparrow \infty$ as $C \uparrow \infty$.
This is because

$$\mathbb{P} \left(\sup_{\mathbf{x} \in \mathcal{N}(n^{-\gamma})^d} |Z_n(\mathbf{x})| > C \cdot \tau_n^{-1} \right) \leq \left(\frac{1}{2} \cdot n^\gamma \right)^d \cdot \sup_{\mathbf{x} \in [0,1]^d} \mathbb{P} \left(|Z_n(\mathbf{x})| > C \cdot \tau_n^{-1} \right).$$

proof

Put

$$\xi_{ni}(\mathbf{x}) = K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) (\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) - \mathbb{E}(\varepsilon_{ni} \cdot \mathbb{I}(|\varepsilon_{ni}| \leq \tau_n) \mid \mathbf{X}_i)).$$

Note that

$$\mathbb{P}(|Z_n(\mathbf{x})| > C \cdot \tau_n^{-1}) = \mathbb{P}(Z_n(\mathbf{x}) > C \cdot \tau_n^{-1}) + \mathbb{P}(-Z_n(\mathbf{x}) > C \cdot \tau_n^{-1}).$$

We find a bound to the first term. The second one can be treated in the same way. We have

$$\begin{aligned} \mathbb{P}(Z_n(\mathbf{x}) > C \cdot \tau_n^{-1}) &= \mathbb{P}\left(\sqrt{\frac{h_1 \dots h_d \log n}{n}} \sum_{i=1}^n \xi_{ni}(\mathbf{x}) > C \log n\right) \\ &\leq n^{-C} \left(\mathbb{E} \exp \left(\sqrt{\frac{h_1 \dots h_d \log n}{n}} \xi_{ni}(\mathbf{x}) \right) \right)^n. \end{aligned}$$

$$\begin{aligned} &\leq n^{-C} \left(\mathbb{E} \exp \left(\sqrt{\frac{h_1 \dots h_d \log n}{n}} \xi_{n1}(\mathbf{x}) \right) \right)^n \\ &\leq n^{-C} \left(1 + e^{2(\sup K)^d} \cdot \frac{h_1 \dots h_d \log n}{2n} \mathbb{E}[\xi_{n1}(\mathbf{x})^2] \right)^n \end{aligned}$$

since $|\sqrt{\frac{h_1 \dots h_d \log n}{n}} \xi_{ni}(\mathbf{x})| \leq 2(\sup K)^d$

$$\leq n^{-C} \exp \left(e^{2(\sup K)^d} \cdot n \left(\frac{h_1 \dots h_d \log n}{2n} \right) \mathbb{E}(\xi_{n1}(\mathbf{x})^2) \right),$$

And $\mathbb{E}[\xi_{ni}(\mathbf{x})^2] \leq \frac{4}{h_1 \dots h_d} \mathbb{E} \left[\prod_{j=1}^d (K^2)_{h_j} (X_{ij} - x_j) \varepsilon_{n1}^2 \right] \leq \frac{C'}{h_1 \dots h_d}$ with large n

Existence of Multivariate LPE

If there exist $d + 1$ points $\mathbf{X}_i$ with $K(\mathbf{X}_i - \mathbf{x}; h) > 0$ that do not lie on a common hyperplane, then $\mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}}$ is invertible so that $\hat{\mathbf{m}}(x)$ exists.

where

$$K(\mathbf{u}; \mathbf{h}) = \prod_{\ell=1}^d K_{h_{\ell}}(u_{\ell}).$$

and $\mathbf{h} = (h_1, \dots, h_d)$

Existence of Multivariate LPE

- A hyperplane in $\mathbb{R}^d$: Set of $\mathbf{u} = (u_1, \dots, u_d)$ such that $\lambda_0 + \lambda_1 u_1 + \dots + \lambda_d u_d = 0$, where at least one of λ_j for $1 \leq j \leq d$ is nonzero.
- Any d or less points in $\mathbb{R}^d$ lie on a common hyperplane.
For $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} \in \mathbb{R}^d$, $\dim \left(\text{span}(\mathbf{x}^{(2)} - \mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)} - \mathbf{x}^{(1)}) \right) \leq k - 1 \leq d - 1$
There is $\lambda = (\lambda_1, \dots, \lambda_d)$ such that at least one of λ_j for $1 \leq j \leq d$ is nonzero and $\lambda^T (\mathbf{x}^{(i)} - \mathbf{x}^{(1)}) = 0$ and set $\lambda_0 = -\lambda^T \mathbf{x}^{(1)}$.
Then we get $\lambda_0 + \lambda^T \mathbf{x}^{(i)} = 0$ for $1 \leq i \leq k$, common hyperplane.
- If a set of d points $\mathbf{x}_1, \dots, \mathbf{x}_d \in \mathbb{R}^d$ are not all on a $(d-2)$ dimensional plane, they determine a unique hyperplane. $\mathcal{H}(\mathbf{x}_1, \dots, \mathbf{x}_d)$

$$\begin{aligned}
0 &= \lambda^\top \mathbf{D}_\mathbf{x}^\top \mathbf{W}_\mathbf{x} \mathbf{D}_\mathbf{x} \lambda \\
&= \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \left[\lambda_0 + \lambda_1 \left(\frac{X_{i,1} - x_1}{h_1} \right) + \cdots + \lambda_d \left(\frac{X_{i,d} - x_d}{h_d} \right) \right]^2
\end{aligned}$$

for some $\lambda \equiv (\lambda_0, \lambda_1, \dots, \lambda_d)^\top$. We show that $\lambda_j = 0$, $1 \leq j \leq d$.

$$\lambda_0 \cdot \mathbf{1} + \lambda_1 \begin{pmatrix} \frac{X_{1,1} - x_1}{h_1} \\ \vdots \\ \frac{X_{d+1,1} - x_1}{h_1} \end{pmatrix} + \cdots + \lambda_d \begin{pmatrix} \frac{X_{1,d} - x_d}{h_d} \\ \vdots \\ \frac{X_{d+1,d} - x_d}{h_d} \end{pmatrix} = 0,$$

Then $\mathbf{u} = \mathbf{X}_i$, for $1 \leq i \leq d+1$ satisfy the following equation:

$$\lambda'_0 + \lambda'_1 \cdot u_1 + \cdots + \lambda'_d \cdot u_d = 0,$$

where $\lambda'_0 = \lambda_0 - \lambda_1 x_1 - \cdots - \lambda_d x_d$ and $\lambda'_j = \lambda_j / h_j$ for $1 \leq j \leq d$.

Since $\mathbf{X}_i$, for $1 \leq i \leq d+1$ do not lie on a common hyperplane, $\lambda = \mathbf{0}$

Existence; fixed $\mathbf{x}$

Let $\mathbf{x} \in [0, 1]^d$ be fixed. Assume that the joint density p of $\mathbf{X}$ is bounded away from zero and infinity on a neighborhood of $\mathbf{x}$ that contains a d -dimensional open rectangle. If $\max_{1 \leq j \leq d} h_j \rightarrow 0$ and $nh_1 \cdots h_d \rightarrow \infty$ as $n \rightarrow \infty$, then

$$\mathbb{P}\left(\mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} \text{ is invertible}\right) \rightarrow 1.$$

Let E_n be the event where $\nexists d + 1$ points $\mathbf{X}_i$ with $K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) > 0$ that do not lie on a common hyperplane in $\mathbb{R}^d$. We prove $\mathbb{P}(E_n) \rightarrow 0$. Let $\mathcal{S}_n$ be the subset of $\{\mathbf{X}_1, \dots, \mathbf{X}_n\}$ such that $\mathbf{X}_i \in \mathcal{S}_n$ if and only if $K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) > 0$.

$$\mathbb{P}(E_n) = \mathbb{P}(|\mathcal{S}_n| \geq d + 1, \text{ any } (d + 1) \text{ points in } \mathcal{S}_n \text{ lie on a common hyperplane}) + \mathbb{P}(|\mathcal{S}_n| \leq d).$$

$$\mathbb{P}(|\mathcal{S}_n| \leq d) = \sum_{j=0}^d \binom{n}{j} q_{\mathbf{x}}^j (1 - q_{\mathbf{x}})^{n-j}$$

$$\leq \exp(-nh_1 \dots h_d \cdot p_{\min, \epsilon}(\mathbf{x})) \cdot (d+1) \cdot \left(2^d nh_1 \dots h_d \cdot p_{\max, \epsilon}(\mathbf{x})\right)^d \text{ for all large } n$$
$$\rightarrow 0$$

using $\binom{n}{i} \leq n^i$, $(1-x)^n \leq \exp(-nx)$ for $x \geq 0$

$$\mathbb{P}(|\mathcal{S}_n| \geq d+1, \text{ any } (d+1) \text{ points in } \mathcal{S}_n \text{ lie on} \\ \text{a common hyperplane}) = \sum_{j=d+1}^n \mathfrak{p}_j$$

where

$$\mathfrak{p}_j := \mathbb{P}(|\mathcal{S}_n| = j, \text{ any } (d+1) \text{ points in } \mathcal{S}_n \text{ lie on a common hyperplane}).$$

For $j \geq d+1$,

$$\begin{aligned} \mathfrak{p}_j &\leq \binom{n}{j} \mathbb{P}(\{\mathbf{X}_1, \dots, \mathbf{X}_j\} = \mathcal{S}_n \text{ and any } (d+1) \text{ points of } \mathcal{S}_n \text{ lie on a common hyperplane}) \\ &\leq \binom{n}{j} \binom{j}{d+1} \cdot \mathbb{P}(\mathbf{X}_i \in [\mathbf{x} - \mathbf{h}, \mathbf{x} + \mathbf{h}] \text{ for } 1 \leq i \leq d \text{ and } \mathbf{X}_{d+1} \in \mathcal{H}(\mathbf{X}_1, \dots, \mathbf{X}_d)). \end{aligned}$$

where $[\mathbf{x} - \mathbf{h}, \mathbf{x} + \mathbf{h}] = \prod_{\ell=1}^d [x_\ell - h_\ell, x_\ell + h_\ell]$

$$\begin{aligned} & \mathbb{P}(\mathbf{X}_i \in [\mathbf{x} - \mathbf{h}, \mathbf{x} + \mathbf{h}] \text{ for } 1 \leq i \leq d \text{ and } \mathbf{X}_{d+1} \in \mathcal{H}(\mathbf{X}_1, \dots, \mathbf{X}_d)) \\ &= \int_{\prod_{i=1}^d [x_i - h_i, x_i + h_i]} \mathbb{P}(\mathbf{X}_{d+1} \in \mathcal{H}(\mathbf{X}_1, \dots, \mathbf{X}_d)) \prod_{i=1}^d p(x_i) dx_1 \cdots dx_d, \end{aligned}$$

and $\mathbb{P}(\mathbf{X}_{d+1} \in \mathcal{H}(\mathbf{X}_1, \dots, \mathbf{X}_d)) = 0$ for all $(\mathbf{X}_1, \dots, \mathbf{X}_d) \in \mathbb{R}^{d \times d}$.

Existence; uniformly

Assume that the joint density p of $\mathbf{X}$ is continuous and bounded away from zero and infinity on $[0, 1]^d$. Suppose that K is Lipschitz continuous and that $\max_{1 \leq j \leq d} h_j \rightarrow 0$ and $nh_1 \cdots h_d / \log n \rightarrow \infty$ as $n \rightarrow \infty$. Then, there exists $\varepsilon > 0$ such that

$$\mathbb{P} \left(\inf_{\mathbf{x} \in [0, 1]^d} \lambda_{\min} \left(n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} \right) > \varepsilon \right) \rightarrow 1,$$

so that $\mathbb{P} \left(\hat{\mathbf{m}}(\mathbf{x}) \text{ is well-defined for all } \mathbf{x} \in [0, 1]^d \right) \rightarrow 1$.

Existence; uniformly

$$\sup_{\mathbf{x} \in [0,1]^d} \left\| n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} - \mathbf{M}(\mathbf{x}) \cdot p(\mathbf{x}) \right\| = o_p(1)$$

where $\mathbf{M}(\mathbf{x})$ is the $(d+1) \times (d+1)$ symmetric matrix with entries

$$[\mathbf{M}(\mathbf{x})]_{0,0} = \prod_{\ell=1}^d \mu_0(x_{\ell}),$$

$$[\mathbf{M}(\mathbf{x})]_{0,j} = \mu_1(x_j) \prod_{\substack{\ell=1 \\ \ell \neq j}}^d \mu_0(x_{\ell}), \quad 1 \leq j \leq d,$$

$$[\mathbf{M}(\mathbf{x})]_{j,j'} = \mu_1(x_j) \mu_1(x_{j'}) \prod_{\substack{\ell=1 \\ \ell \neq j, j'}}^d \mu_0(x_{\ell}), \quad 1 \leq j \neq j' \leq d,$$

$$[\mathbf{M}(\mathbf{x})]_{j,j} = \mu_2(x_j) \prod_{\ell=1, \ell \neq j}^d \mu_0(x_{\ell}), \quad 1 \leq j \leq d.$$

Existence; uniformly

since

$$[n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}}]_{0,0} = \sum_{m=1}^n \prod_{l=1}^d K_{h_l}(X_{m,l} - x_l)$$

$$[n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}}]_{0,j} = \sum_{m=1}^n \left(\frac{X_{m,j} - x_j}{h_j} \right) \prod_{l=1}^d K_{h_l}(X_{m,l} - x_l), \quad 1 \leq j \leq d$$

$$[n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}}]_{j,j'} = \sum_{m=1}^n \left(\frac{X_{m,j} - x_j}{h_j} \right) \left(\frac{X_{m,j'} - x_{j'}}{h_{j'}} \right) \prod_{l=1}^d K_{h_l}(X_{m,l} - x_l), \quad 1 \leq j \neq j' \leq d,$$

$$[n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}}]_{j,j} = \sum_{m=1}^n \left(\frac{X_{m,j} - x_j}{h_j} \right)^2 \prod_{l=1}^d K_{h_l}(X_{m,l} - x_l), \quad 1 \leq j \leq d$$

Existence; uniformly

If $p(\mathbf{x})$ is continuous,

$$\begin{aligned} & \sup_{\mathbf{x} \in [0,1]^d} \left\| \sum_{m=1}^n \left(\frac{X_{m,j} - x_j}{h_j} \right) \left(\frac{X_{m,j'} - x'_{j'}}{h'_{j'}} \right) \prod_{l=1}^d K_{h_l}(X_{m,l} - x_l) \right. \\ & \quad \left. - \mathbb{E} \left[\left(\frac{X_{1,j} - x_j}{h_j} \right) \left(\frac{X_{1,j'} - x'_{j'}}{h'_{j'}} \right) \prod_{l=1}^d K_{h_l}(X_{1,l} - x_l) \right] \right\| = O_p\left(\frac{\sqrt{\log n}}{\sqrt{nh_1 \dots h_d}}\right) = o_p(1) \\ & \quad \sup_{\mathbf{x} \in [0,1]^d} \left\| \mathbb{E} \left[\left(\frac{X_{1,j} - x_j}{h_j} \right) \left(\frac{X_{1,j'} - x'_{j'}}{h'_{j'}} \right) \prod_{l=1}^d K_{h_l}(X_{1,l} - x_l) \right] \right. \\ & \quad \left. - p(\mathbf{x}) \cdot \mu_1(x_j) \mu_1(x_{j'}) \prod_{\substack{\ell=1 \\ \ell \neq j, j'}}^d \mu_0(x_\ell) \right\| = o_p(1) \end{aligned}$$

Existence; uniformly

$$\xi_{ni}(\mathbf{x}) = \left(\frac{X_{i,j} - x_j}{h_j} \right) \left(\frac{X_{i,j'} - x'_j}{h'_j} \right) \prod_{l=1}^d K_{h_l}(X_{i,l} - x_l) - p(\mathbf{x}) \mu_1(x_j) \mu_1(x'_j) \prod_{\substack{\ell=1 \\ \ell \neq j, j'}}^d \mu_0(x_\ell)$$

$$\mathbb{E}[\xi_{ni}(\mathbf{x})] = \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} t_i t_{j'} K(t_1) \dots K(t_d) \{p(\mathbf{x}) + (p(\mathbf{x} + \mathbf{h}\mathbf{t}) - p(\mathbf{x}))\} dt_1 \dots dt_d$$

$$-p(\mathbf{x}) \mu_1(x_j) \mu_1(x'_j) \prod_{\substack{\ell=1 \\ \ell \neq j, j'}}^d \mu_0(x_\ell)$$

$$\sup_{\mathbf{x} \in [0,1]^d} \int \dots \int t_i t_{j'} K(t_1) \dots K(t_d) \{(p(\mathbf{x} + \mathbf{h}\mathbf{t}) - p(\mathbf{x}))\} dt_1 \dots dt_d$$

Existence; uniformly

Let's prove that

$$\inf_{\mathbf{x} \in [\mathbf{h}, \mathbf{1} - \mathbf{h}]} \lambda_{\min}(\mathbf{M}(\mathbf{x})) > 0$$

$$\text{For } \mathbf{x} \in [\mathbf{h}, \mathbf{1} - \mathbf{h}], \mathbf{M}(\mathbf{x}) \equiv \mathbf{M} = \begin{pmatrix} \mu_0^d & \mu_1 \mu_0^{d-1} & \mu_1 \mu_0^{d-1} \dots \\ & \mu_2 \mu_0^{d-2} & \mu_1^2 \mu_0^{d-2} \dots \\ & & \dots \end{pmatrix}.$$

Thus,

$$\inf_{\mathbf{x} \in [\mathbf{h}, \mathbf{1} - \mathbf{h}]} \lambda_{\min}(\mathbf{M}(\mathbf{x})) = \lambda_{\min}(\mathbf{M}).$$

Now, note that

$$\begin{aligned} \boldsymbol{\alpha}^\top \mathbf{M} \boldsymbol{\alpha} &= \alpha_0^2 \mu_0^d + (\alpha_1^2 + \dots + \alpha_d^2) (\mu_2 \mu_0^{d-2}) + \sum_{i \neq j} \alpha_i \alpha_j \mu_1^2 \mu_0^{d-2} + 2 \sum_i \alpha_0 \alpha_i \mu_1 \mu_0^{d-1} \\ &= \int_{[-1,1]^d} (\alpha_0 + \alpha_1 u_1 + \dots + \alpha_d u_d)^2 \prod_{j=1}^d K(u_j) du_1 \dots du_d. \end{aligned}$$

Existence; uniformly

Suppose that $\alpha^\top \mathbf{M} \alpha = 0$ for some α . Then,

$$\text{Leb}_d \left(\left\{ \mathbf{u} \in [-1, 1]^d : \alpha_0 + \alpha_1 u_1 + \cdots + \alpha_d u_d \neq 0 \right\} \right) = 0,$$

which implies that almost all $\mathbf{u}$ in $[-1, 1]^d$ satisfy the equation

$$\alpha_0 + \alpha_1 u_1 + \cdots + \alpha_d u_d = 0.$$

This is obviously false when at least one of α_j for $1 \leq j \leq d$ is nonzero since in that case the equation

$$\alpha_0 + \alpha_1 u_1 + \cdots + \alpha_d u_d = 0$$

defines a hyperplane. When all α_j for $1 \leq j \leq d$ are zero, then

$$0 = \alpha^\top \mathbf{M} \alpha = \alpha_0^2 \cdot \int_{[-1, 1]^d} \prod_{j=1}^d K(u_j) du_1 \cdots du_d = \alpha_0^2.$$

Existence; uniformly

Let's prove that

$$\inf_{\mathbf{x} \in [0,1]^d \setminus [\mathbf{h}, \mathbf{1}-\mathbf{h}]} \lambda_{\min}(\mathbf{M}(\mathbf{x})) > 0.$$

$$\mu_j(x_i) = \int_{-\beta}^1 v^j K(v) \text{ or } \int_{-1}^{\beta} v^j K(v) \text{ or } \int_{-1}^1 v^j K(v), \quad 1 \leq i \leq d$$

Then we get $\mathbf{M}_l(\beta)$ for $1 \leq l \leq 3^d - 1$ types of matrices.

$$\inf_{\mathbf{x} \in [0,1]^d \setminus [\mathbf{h}, \mathbf{1}-\mathbf{h}]} \lambda_{\min}(\mathbf{M}(\mathbf{x})) = \inf_{\beta_1, \dots, \beta_k \in [0,1]} \min_{1 \leq l \leq 3^d - 1} \lambda_{\min}(\mathbf{M}_l(\beta)) > 0$$

since $\lambda_{\min}(\mathbf{M}_l(\beta)) > 0$, continuous with respect to β_i

Existence; uniformly

$$\mathbf{M}(\mathbf{x}) = \begin{pmatrix} (\int K(v))^d & (\int vK(v))(\int K(v))^{d-1} \dots & \\ & (\int v^2 K(v))(\int K(v))^{d-2} & \dots \\ & & \dots \end{pmatrix}$$

Using $|\lambda_{\min}(A) - \lambda_{\min}(B)|^2 \leq \text{tr}[(A - B)^2]$, $f(\beta) := \lambda_{\min}(\mathbf{M}_l(\beta))$

$$|f(\beta_j + \delta) - f(\beta_j)|^2 \leq \text{tr} [\mathbf{M}_l(\beta_j + \delta) - \mathbf{M}_l(\beta_j)] \rightarrow 0 \text{ as } \delta \rightarrow 0$$

since $\int_{-\beta-\delta}^{-\beta} v^j K(v) \rightarrow 0$ as $\delta \rightarrow 0$

$$\begin{aligned} \boldsymbol{\alpha}^\top \mathbf{M}_l(\beta) \boldsymbol{\alpha} &= \alpha_0^2 \prod \mu_0(x_l) + \alpha_1^2 \mu_2(x_1) \prod_{l \neq 1} \mu_0(x_l) + \dots + \alpha_d^2 \mu_2(x_d) \prod_{l \neq d} \mu_0(x_l) \\ &+ \sum_{i \neq j} \alpha_i \alpha_j \mu_1(x_i) \mu_j(x_j) \prod_{l \neq i, j} \mu_0(x_l) + 2 \sum_i \alpha_0 \alpha_i \mu_1(x_i) \prod_{l=1}^d \mu_0(x_l) \end{aligned}$$

Existence; uniformly

$$= \int_{[-1,1]^d} (\alpha_0 + \alpha_1 u_1 + \cdots + \alpha_d u_d)^2 \prod_{j=1}^d K(u_j) du_1 \cdots du_d = 0$$

$$\text{Leb}_d \left(\left\{ \mathbf{u} \in [-1, 1]^d : \alpha_0 + \alpha_1 u_1 + \cdots + \alpha_d u_d \neq 0 \right\} \right) = 0,$$

Existence; uniformly

Now we get

$$\sup_{\mathbf{x} \in [0,1]^d} \left\| n^{-1} \cdot \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} - \mathbf{M}(\mathbf{x}) \cdot p(\mathbf{x}) \right\| = o_p(1)$$

$$\inf_{\mathbf{x} \in [0,1]^d} \lambda_{\min}(\mathbf{M}(\mathbf{x})) > 0.$$

Using $|\lambda_{\min}(A) - \lambda_{\min}(B)|^2 \leq \text{tr}[(A - B)^2]$,

$$\inf_{\mathbf{x} \in [0,1]^d} \lambda_{\min} \left(n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} \right) > 0$$

with probability tending to 1.

Pointwise error; Notation

Put $\mathbf{m}(\mathbf{x}) := (m(\mathbf{x}), h_1 \nabla_1(m)(\mathbf{x}), \dots, h_d \nabla_d(m)(\mathbf{x}))^\top$. Let

$$\nabla_{\mathbf{r}}(m)(\mathbf{x}) = \frac{\partial^{|\mathbf{r}|}}{\partial x_1^{r_1} \dots \partial x_d^{r_d}}(m)(\mathbf{x}).$$

For a d -tuple of nonnegative integers $\mathbf{r} \equiv (r_1, \dots, r_d)$, let

$|\mathbf{r}| = r_1 + \dots + r_d$, $\mathbf{a}^{\mathbf{r}} = a_1^{r_1} \times \dots \times a_d^{r_d}$ for $\mathbf{a} = (a_1, \dots, a_d) \in \mathbb{R}^d$ and

$\mathbf{r}! = r_1! \dots r_d!$. Define

$$\tilde{\mu}_{\mathbf{r}}(\mathbf{x}) = \int_{[0,1]^d} \left(\frac{\mathbf{u} - \mathbf{x}}{\mathbf{h}} \right)^{\mathbf{r}} \begin{pmatrix} 1 \\ \frac{\mathbf{u} - \mathbf{x}}{\mathbf{h}} \end{pmatrix} K(\mathbf{u} - \mathbf{x}; \mathbf{h}) d\mathbf{u} \in \mathbb{R}^{d+1}.$$

Note that for a symmetric K and $\mathbf{x} \in [\mathbf{h}, \mathbf{1} - \mathbf{h}]$, we have

$$\tilde{\mu}_{(1,1,0,\dots,0)}(\mathbf{x}) \equiv \tilde{\mu}_{(1,0,1,\dots,0)}(\mathbf{x}) \equiv \dots \equiv \tilde{\mu}_{(0,0,\dots,1,1)}(\mathbf{x}) \equiv \mathbf{0},$$

$$\tilde{\mu}_{(2,0,0,\dots,0)}(\mathbf{x}) \equiv \tilde{\mu}_{(0,2,0,\dots,0)}(\mathbf{x}) \equiv \dots \equiv \tilde{\mu}_{(0,0,\dots,0,2)}(\mathbf{x}) \equiv (\mu_2, 0, \dots, 0)^\top.$$

Pointwise error

Theorem 10.4

Assume that m is twice partially continuously differentiable at $\mathbf{x}$ and that p is continuous at $\mathbf{x}$ and $p(\mathbf{x}) > 0$. If $\max_{1 \leq j \leq d} h_j \rightarrow 0$ and $nh_1 \cdots h_d \rightarrow \infty$ as $n \rightarrow \infty$, then

$$\begin{aligned} & \mathbb{E}(\hat{m}(\mathbf{x}) | \mathbf{X}_1, \dots, \mathbf{X}_n) - m(\mathbf{x}) \\ &= \sum_{\mathbf{r}: |\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}} \cdot \frac{1}{\mathbf{r}!} \cdot \mathbf{M}(\mathbf{x})^{-1} \tilde{\boldsymbol{\mu}}_{\mathbf{r}}(\mathbf{x}) \cdot \nabla_{\mathbf{r}}(m)(\mathbf{x}) + o_p\left(\sum_{\mathbf{r}: |\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}}\right). \end{aligned}$$

Pointwise error

$$E(\hat{\mathbf{m}}(\mathbf{x}) \mid X_n) - \mathbf{m}(\mathbf{x}) = \left(n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} \mathbf{D}_{\mathbf{x}} \right)^{-1} n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{m} - \mathbf{D}_{\mathbf{x}} \mathbf{m}(\mathbf{x}))$$

$$\hat{\beta}_{j,l}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \left(\frac{X_{i,j} - x_j}{h_j} \right)^l \{ m(\mathbf{X}_i) - m(\mathbf{x}) - (\mathbf{X}_i - \mathbf{x})^T \nabla m(\mathbf{x}) \}$$

$$\left[n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{m} - \mathbf{D}_{\mathbf{x}} \mathbf{m}(\mathbf{x})) \right]_j = \hat{\beta}_{j,0}(\mathbf{x}), \quad j = 0$$

$$\left[n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{m} - \mathbf{D}_{\mathbf{x}} \mathbf{m}(\mathbf{x})) \right]_j = \hat{\beta}_{j,1}(\mathbf{x}), \quad 1 \leq j \leq d$$

Pointwise error

Use

$$m(\mathbf{x} + \mathbf{y}) = \sum_{k=0}^r \frac{1}{k!} D^k m(\mathbf{x})[\mathbf{y}^k] + \frac{1}{(r-1)!} \int_0^1 (1-w)^{r-1} (D^r m(\mathbf{x} + \mathbf{y}w) - D^r m(\mathbf{x})[(\mathbf{y})^r]) dw$$

where

$$D^r m(\mathbf{x})[\mathbf{y}^r] = \sum_{i_1, \dots, i_r=1}^d \frac{\partial^r m}{\partial x_{i_1} \dots \partial x_{i_r}}(\mathbf{x}) y_{i_1} \dots y_{i_r}$$

Pointwise error

$$\begin{aligned}
 E\hat{\beta}_{j,l}(\mathbf{x}) &= E \left[K(\mathbf{X}_1 - \mathbf{x}; \mathbf{h}) \left(\frac{X_{1,j} - x_j}{h_j} \right)^l \{ m(\mathbf{X}_1) - m(\mathbf{x}) - (\mathbf{X}_1 - \mathbf{x})^T \nabla m(\mathbf{x}) \} \right] \\
 &= \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K(t_1) \dots K(t_d) (t_j)^l \left\{ \frac{1}{2} D^2 m(\mathbf{x}) [(\mathbf{h}\mathbf{t})^2] + \right. \\
 &\quad \left. \int_0^1 (1-w) (D^2 m(\mathbf{x} + \mathbf{h}\mathbf{t}w) - D^2 m(\mathbf{x})) [(\mathbf{h}\mathbf{t})^2] dw \right\} \\
 &\quad \cdot \{ p(\mathbf{x}) + (p(\mathbf{x} + \mathbf{h}\mathbf{t}) - p(\mathbf{x})) \} dt_1 \dots dt_d \\
 &= p(\mathbf{x}) \sum_{\mathbf{r}: |\mathbf{r}|=2} \left(\mathbf{h}^{\mathbf{r}} \frac{1}{\mathbf{r}!} \mu_{r_j+l}(x_j) \prod_{l \neq j} \mu_{r_l}(x_l) \cdot \nabla_{\mathbf{r}}(m)(\mathbf{x}) \right) + o\left(\sum_{\mathbf{r}: |\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}} \right)
 \end{aligned}$$

Pointwise error

$$\begin{aligned} Var \hat{\beta}_{j,l}(\mathbf{x}) &= \frac{1}{nh_1 \dots h_d} \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K^2(t_1) \dots K^2(t_d) (t_j)^{2l} \left\{ \frac{1}{2} D^2 m(\mathbf{x}) [(\mathbf{h}\mathbf{t})^2] + \right. \\ &\quad \left. \int_0^1 (1-w) (D^2 m(\mathbf{x} + \mathbf{h}\mathbf{t}w) - D^2 m(\mathbf{x})) [(\mathbf{h}\mathbf{t})^2] dw \right\}^2 \\ &\quad \cdot \{p(\mathbf{x}) + (p(\mathbf{x} + \mathbf{h}\mathbf{t}) - p(\mathbf{x}))\}^2 dt_1 \dots dt_d - \frac{1}{n} (E \hat{\beta}_{j,l}(\mathbf{x}))^2 \\ &= O\left(\frac{1}{nh_1 \dots h_d} \left(\sum_{\mathbf{r}:|\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}}\right)^2\right) = o\left(\sum_{\mathbf{r}:|\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}}\right)^2 \end{aligned}$$

Pointwise error

Therefore,

$$\hat{\beta}_{j,l}(\mathbf{x}) = p(\mathbf{x}) \sum_{\mathbf{r}:|\mathbf{r}|=2} \left(\mathbf{h}^{\mathbf{r}} \frac{1}{\mathbf{r}!} \mu_{r_j+l}(x_j) \prod_{l \neq j} \mu_{r_l}(x_l) \cdot \nabla_{\mathbf{r}}(m)(\mathbf{x}) \right) + o_p \left(\sum_{\mathbf{r}:|\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}} \right)$$

Recall that

$$\tilde{\mu}_r(\mathbf{x}) = \int_{[0,1]^d} \left(\frac{\mathbf{u} - \mathbf{x}}{\mathbf{h}} \right)^r \left(\frac{1}{\mathbf{h}} \right) K(\mathbf{u} - \mathbf{x}; \mathbf{h}) d\mathbf{u}$$

$$\left[n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{m} - \mathbf{D}_{\mathbf{x}} \mathbf{m}(\mathbf{x})) \right]_j = \hat{\beta}_{j,0}(\mathbf{x}), \quad j = 0$$

$$\left[n^{-1} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{m} - \mathbf{D}_{\mathbf{x}} \mathbf{m}(\mathbf{x})) \right]_j = \hat{\beta}_{j,1}(\mathbf{x}), \quad 1 \leq j \leq d$$

Now we get

$$E(\hat{\mathbf{m}}(\mathbf{x}) \mid X_n) - \mathbf{m}(\mathbf{x}) = \sum_{\mathbf{r}:|\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}} \cdot \frac{1}{\mathbf{r}!} \cdot \mathbf{M}(\mathbf{x})^{-1} \tilde{\mu}_{\mathbf{r}}(\mathbf{x}) \cdot \nabla_{\mathbf{r}}(m)(\mathbf{x}) + o_p \left(\sum_{\mathbf{r}:|\mathbf{r}|=2} \mathbf{h}^{\mathbf{r}} \right).$$

Pointwise error

Assume that $\sigma^2(\cdot) = \text{var}(Y \mid \mathbf{X} = \cdot)$ and p are continuous at $\mathbf{x}$ and $p(\mathbf{x}) > 0$. If $\max_{1 \leq j \leq d} h_j \rightarrow 0$ and $nh_1 \cdots h_d \rightarrow \infty$ as $n \rightarrow \infty$, then

$$\text{var}(\hat{\mathbf{m}}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n) = n^{-1} h_1^{-1} \cdots h_d^{-1} \cdot \mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} \cdot \frac{\sigma^2(\mathbf{x})}{p(\mathbf{x})} + o_p(n^{-1} h_1^{-1} \cdots h_d^{-1}).$$

Pointwise error

$$\text{var}(\hat{\mathbf{m}}(\mathbf{x}) \mid X_1, \dots, X_n) = \left(\mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \left(\mathbf{D}_x^\top \mathbf{W}_x \mathbf{D}_x \right)^{-1}$$

where $\widetilde{\mathbf{W}}_x := \mathbf{W}_x \text{diag}(\sigma^2(X_i)) \mathbf{W}_x = \text{diag} \left((K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}))^2 \sigma^2(X_i) \right)$.

$$\left[n^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \right]_{i,j} = \frac{1}{n} \sum_{l=1}^n (K(\mathbf{X}_l - \mathbf{x}; \mathbf{h}))^2 \sigma^2(X_l) \left(\frac{X_{l,i} - x_i}{h_i} \right) \left(\frac{X_{l,j} - x_j}{h_j} \right), \quad i, j \geq 1$$

$$\begin{aligned} & \mathbb{E} \left((K(\mathbf{X}_1 - \mathbf{x}; \mathbf{h}))^2 \sigma^2(\mathbf{X}_1) \left(\frac{X_{1,i} - x_i}{h_i} \right) \left(\frac{X_{1,j} - x_j}{h_j} \right) \right) \\ &= \frac{1}{h_1 \dots h_d} \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K^2(t_1) \dots K^2(t_d) (t_i) (t_j) \sigma^2(\mathbf{x} + \mathbf{h}\mathbf{t}) p(\mathbf{x} + \mathbf{h}\mathbf{t}) dt_1 \dots dt_d \\ &= \frac{1}{h_1 \dots h_d} \kappa_1(x_i) \kappa_1(x_j) \prod_{l \neq i, j} \kappa_0(x_l) \sigma^2(\mathbf{x}) p(\mathbf{x}) + o\left(\frac{1}{h_1 \dots h_d}\right) \end{aligned}$$

Pointwise error

$$\begin{aligned} \text{Var} \left[n^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x \right]_{i,j} &= \frac{1}{n} \text{Var} (K(\mathbf{X}_1 - \mathbf{x}; \mathbf{h}))^2 \sigma^2(\mathbf{X}_1) \left(\frac{X_{1,i} - x_i}{h_i} \right) \left(\frac{X_{1,j} - x_j}{h_j} \right) \\ &= O\left(\frac{1}{nh_1^2 \dots h_d^2}\right) = o\left(\frac{1}{h_1^2 \dots h_d^2}\right) \end{aligned}$$

Therefore,

$$n^{-1} \mathbf{D}_x^\top \widetilde{\mathbf{W}}_x \mathbf{D}_x = \frac{1}{h_1 \dots h_d} \sigma^2(\mathbf{x}) p(\mathbf{x}) \mathbf{S}(\mathbf{x}) + o_p\left(\frac{1}{h_1 \dots h_d}\right)$$

where $\mathbf{S}(\mathbf{x})$ is the $(d+1) \times (d+1)$ symmetric matrix with entries

$$[\mathbf{S}(\mathbf{x})]_{0,0} = \prod_{\ell=1}^d \kappa_0(x_\ell), [\mathbf{S}(\mathbf{x})]_{0,j} = \kappa_1(x_j) \prod_{\substack{\ell=1 \\ \ell \neq j}}^d \kappa_0(x_\ell), \quad 1 \leq j \leq d,$$

$$[\mathbf{S}(\mathbf{x})]_{j,j'} = \kappa_1(x_j) \kappa_1(x_{j'}) \prod_{\ell=1}^d \kappa_0(x_\ell), [\mathbf{S}(\mathbf{x})]_{j,j} = \kappa_2(x_j) \prod_{\ell=1}^d \kappa_0(x_\ell)$$

Pointwise error

Let's prove that

$$\|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\| = O_p \left(\max_j h_j^2 + n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \right).$$

Using

$$\begin{aligned} \mathbb{E} \left(\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) &= (\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x))^T (\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x)) \\ &\quad + \text{tr}(\text{var}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n)) \\ &\leq C_1 \max_j h_j^4 (1 + o_p(1)) + \frac{C_2}{n h_1 \dots h_d} (1 + o_p(1)) \end{aligned}$$

Pointwise error

$$Z_n = \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2$$

$$E \left[\frac{|Z_n|}{(\max_j h_j^2 + \frac{1}{\sqrt{nh_1 \dots h_d}})^2} \mid \mathbb{X}_n \right] \leq 2\max\{C_1, C_2\} + o_p(1)$$

$$P \left(E \left[\frac{|Z_n|}{(\max_j h_j^2 + \frac{1}{\sqrt{nh_1 \dots h_d}})^2} \mid \mathbb{X}_n \right] < C + 1 \right) \rightarrow 1$$

$\mathcal{E}_n : D_x^T W_w D_x$ is invertible, $P(\mathcal{E}_n) \rightarrow 1$ Therefore,

$$\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\| = O_p(\max_j h_j^2 + n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2})$$

Assume that m has bounded second-order partial derivatives and that the joint density p of $\mathbf{X}$ is bounded away from zero and infinity on $[0, 1]^d$. Suppose that K is Lipschitz continuous and that $\max_{1 \leq j \leq d} h_j \rightarrow 0$ and $nh_1 \cdots h_d / \log n \rightarrow \infty$ as $n \rightarrow \infty$. Then,

$$\mathbb{E} \left(\int_{[0,1]^d} \|\hat{m}(\mathbf{x}) - m(\mathbf{x})\|^2 d\mathbf{x} \mid \mathbf{X}_1, \dots, \mathbf{X}_n \right) = O_p \left(\max_{1 \leq j \leq d} h_j^4 + n^{-1} h_1^{-1} \cdots h_d^{-1} \right).$$

$$\begin{aligned}
 \sup_{\mathbf{x} \in [0,1]^d} \mathbb{E} \left(\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) &\leq \sup_{\mathbf{x} \in [0,1]^d} \left(\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x) \right)^T \left(\mathbb{E}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) - \mathbf{m}(x) \right) \\
 &\quad + \sup_{\mathbf{x} \in [0,1]^d} \text{tr} \left(\text{var}(\hat{\mathbf{m}}(x) \mid \mathbb{X}_n) \right) \\
 &\leq \hat{\lambda}_{\inf}^{-2} \cdot \sup_{\mathbf{x} \in [0,1]^d} \left(\hat{\beta}_{0,1}(x)^2 + \hat{\beta}_{1,1}(x)^2 + \dots + \hat{\beta}_{d,1}(x)^2 \right) \\
 &\quad + \hat{\lambda}_{\inf}^{-2} \cdot \sup_{\mathbf{x} \in [0,1]^d} \left(\hat{\gamma}_{0,2}(x) + \hat{\gamma}_{1,2}(x) + \dots + \hat{\gamma}_{d,2}(x) \right) \cdot n^{-1} h_1^{-1} \dots h_d^{-1}
 \end{aligned}$$

where

$$\hat{\gamma}_{j,\ell}(\mathbf{x}) = n^{-1} \sum_{i=1}^n \left(\frac{X_{i,j} - x_j}{h_j} \right)^\ell \left(K^2(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \cdot \sigma^2(\mathbf{X}_i) \right).$$

L^2 error

Recall

$$\hat{\beta}_{j,l}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \left(\frac{X_{i,j} - x_j}{h_j} \right)^l \{m(\mathbf{X}_i) - m(\mathbf{x}) - (\mathbf{X}_i - \mathbf{x})^T \nabla m(\mathbf{x})\}$$

With probability tending to 1,

$$\sup_{\mathbf{x} \in [0,1]^d} \left| \mathbb{E}(\hat{\beta}_{j,\ell}(\mathbf{x})) \right| \leq C_1 \cdot \max_j h_j^2$$

since

$$\begin{aligned} \sup_{\mathbf{x} \in [0,1]^d} \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K(t_1) \dots K(t_d) (t_j)^l \left\{ \frac{1}{2} D^2 m(\tilde{\mathbf{x}}) [(\mathbf{h}\mathbf{t})^2] \right\} p(\mathbf{x} + \mathbf{h}\mathbf{t}) dt_1 \dots dt_d \\ \leq C_1 \cdot \max_j h_j^2 \end{aligned}$$

for some constant $0 < C_1 < \infty$, and

$$\sup_{\mathbf{x} \in [0,1]^d} \left| \hat{\beta}_{j,\ell}(\mathbf{x}) - \mathbb{E}(\hat{\beta}_{j,\ell}(\mathbf{x})) \right| = O_p \left(n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \sqrt{\log n} \cdot \max_j h_j^2 \right) = o_p \left(\max_j h_j^2 \right).$$

L^2 error

Recall

$$\hat{\gamma}_{j,\ell}(\mathbf{x}) = n^{-1} \sum_{i=1}^n \left(\frac{X_{i,j} - x_j}{h_j} \right)^\ell \left(K^2(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \cdot \sigma^2(\mathbf{X}_i) \right).$$

With probability tending to 1,

$$\sup_{\mathbf{x} \in [0,1]^d} |\mathbb{E} \hat{\gamma}_{j,\ell}(\mathbf{x})| \leq C$$

since

$$\sup_{\mathbf{x} \in [0,1]^d} \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K^2(t_1) \dots K^2(t_d) (t_j)^\ell \sigma^2(\mathbf{x} + \mathbf{h}\mathbf{t}) p(\mathbf{x} + \mathbf{h}\mathbf{t}) dt_1 \dots dt_d \leq C_2$$

for some constant $0 < C_2 < \infty$, and

$$\sup_{\mathbf{x} \in [0,1]^d} |\hat{\gamma}_{j,\ell}(\mathbf{x}) - \mathbb{E}(\hat{\gamma}_{j,\ell}(\mathbf{x}))| = O_p \left(n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \sqrt{\log n} \right) = o_p(1)$$

L^2 error

Therefore,

$$\begin{aligned} \sup_{\mathbf{x} \in [0,1]^d} \mathbb{E} \left(\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 \mid \mathbb{X}_n \right) &\leq \hat{\lambda}_{\inf}^{-2} \cdot \sup_{\mathbf{x} \in [0,1]^d} \left(\hat{\beta}_{0,1}(x)^2 + \hat{\beta}_{1,1}(x)^2 + \dots + \hat{\beta}_{d,1}(x)^2 \right) \\ &\quad + \hat{\lambda}_{\inf}^{-2} \cdot \sup_{\mathbf{x} \in [0,1]^d} (\hat{\gamma}_{0,2}(x) + \hat{\gamma}_{1,2}(x) + \dots + \hat{\gamma}_{d,2}(x)) \cdot n^{-1} h_1^{-1} \dots h_d^{-1} \\ &\leq C' (\max h_j^4 + \frac{1}{n h_1 \dots h_d}) \end{aligned}$$

with probability tending to 1, which means that

$$\mathbb{E} \left(\int_{[0,1]^d} \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\|^2 d\mathbf{x} \mid \mathbf{X}_1, \dots, \mathbf{X}_n \right) = O_p \left(\max_{1 \leq j \leq d} h_j^4 + n^{-1} h_1^{-1} \dots h_d^{-1} \right).$$

L^2 error

We have proved that

$$\mathbb{E} \left(\int_{[0,1]^d} \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\|^2 d\mathbf{x} \mid \mathbf{X}_1, \dots, \mathbf{X}_n \right) = O_p \left(\max_{1 \leq j \leq d} h_j^4 + \frac{1}{nh_1 \dots h_d} \right).$$

for some $C' < \infty$, with probability tending to 1.

Let $Z_n = \int_0^1 \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\|^2 dx$, Then

$$P \left(E \left[\frac{|Z_n|}{(\max_{1 \leq j \leq d} h_j^4 + \frac{1}{nh_1 \dots h_d})} \mid \mathbb{X}_n \right] \leq C' \right) \rightarrow 1$$

$\mathcal{E}_n : \inf_{x \in [0,1]} \lambda_{\min}(\frac{1}{n} D_x^T W_x D_x) > \epsilon$, $P(\mathcal{E}_n) \rightarrow 1$ Therefore,

$$\int_{[0,1]^d} \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\|^2 dx = O_p(\max h_j^4 + \frac{1}{nh_1 \dots h_d})$$

L^∞ error

Assume that $\mathbb{E}[\exp(\alpha|Y - m(\mathbf{X})|)] < \infty$ for some $\alpha > 0$, $\mathbb{E}(Y^2 \mid \mathbf{X} = \cdot)$ is bounded on $[0, 1]^d$ and p is bounded away from zero and infinity on $[0, 1]^d$. Suppose that K is Lipschitz continuous and that $(nh_1 \cdots h_d)/(\log n)^3 \rightarrow \infty$ as $n \rightarrow \infty$. Then, it holds that

$$\sup_{\mathbf{x} \in [0,1]^d} \|\hat{m}(\mathbf{x}) - \mathbb{E}(\hat{m}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n)\| = O_p \left(n^{-1/2} h_1^{-1/2} \cdots h_d^{-1/2} \sqrt{\log n} \right).$$

Therefore,

$$\begin{aligned} \sup_{\mathbf{x} \in [0,1]^d} \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbf{m}(\mathbf{x})\| &\leq \sup_{\mathbf{x} \in [0,1]^d} \|\mathbb{E}(\hat{\mathbf{m}}(\mathbf{x}) \mid \mathbb{X}_n) - \mathbf{m}(\mathbf{x})\| + \sup_{\mathbf{x} \in [0,1]^d} \|\hat{\mathbf{m}}(\mathbf{x}) - \mathbb{E}(\hat{\mathbf{m}}(\mathbf{x}) \mid \mathbb{X}_n)\| \\ &= O_p \left(\max_j h_j^2 + n^{-1/2} h_1^{-1/2} \cdots h_d^{-1/2} \sqrt{\log n} \right). \end{aligned}$$

Asymptotic normality

Let $\mathbf{x} \in [0, 1]^d$ be fixed. Assume that $\mathbb{E}(|Y - m(\mathbf{X})|^{2+\delta}) < \infty$ for some $\delta > 0$, that σ^2 and p are continuous at $\mathbf{x}$ and that $p(\mathbf{x}) > 0$. If $\max_j h_j \rightarrow 0$ and $nh_1 \cdots h_d \rightarrow \infty$ as $n \rightarrow \infty$, then

$$\text{var}(\hat{m}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n)^{-1/2} (\hat{m}(\mathbf{x}) - \mathbb{E}(\hat{m}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n)) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}_{d+1}),$$

where $\mathbf{I}_{d+1}$ denotes the $(d+1) \times (d+1)$ identity matrix.

Asymptotic normality

$$\begin{aligned} & \text{var}(\hat{\mathbf{m}}(x)|\mathbb{X}_n)^{-1/2} (\hat{\mathbf{m}}(x) - E(\hat{\mathbf{m}}(x)|\mathbb{X}_n)) \\ &= \left[\frac{\sigma^2(\mathbf{x})p(\mathbf{x})}{nh_1 \dots h_d} \cdot \mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} + o_p\left(\frac{1}{nh_1 \dots h_d}\right) \right]^{-1/2} \\ & [p(\mathbf{x})\mathbf{M}(\mathbf{x}) + o_p(1)]^{-1} \cdot \frac{1}{n} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (Y - m) = \mathbf{Z}_n(\mathbf{x})(1 + o_p(1)) \end{aligned}$$

where

$$\mathbf{Z}_n(\mathbf{x}) = \left[\frac{\sigma^2(\mathbf{x})p(\mathbf{x})}{nh_1 \dots h_d} \cdot \mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} \right]^{-1/2} [p(\mathbf{x})\mathbf{M}(\mathbf{x})]^{-1} \frac{1}{n} \mathbf{D}_{\mathbf{x}}^{\top} \mathbf{W}_{\mathbf{x}} (\mathbf{Y} - \mathbf{m})$$

Asymptotic normality

For $a \in \mathbb{R}^{d+1}$, define $\zeta_{ni}(\mathbf{x})$ by $a^\top Z_n(\mathbf{x}) = \sum_{i=1}^n \zeta_{ni}(\mathbf{x})$,

$$b(\mathbf{x})^T := a^T \left[\frac{\sigma^2(\mathbf{x})p(\mathbf{x})}{nh_1 \dots h_d} \cdot \mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} \right]^{-1/2} [p(\mathbf{x})\mathbf{M}(\mathbf{x})]^{-1}$$

$$\left[\mathbf{D}_{\mathbf{x}}^T \mathbf{W}_{\mathbf{x}}(\mathbf{Y} - \mathbf{m}) \right]_0 = \left(\sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h})(Y_i - m(\mathbf{X}_i)) \right)$$

$$\left[\mathbf{D}_{\mathbf{x}}^T \mathbf{W}_{\mathbf{x}}(\mathbf{Y} - \mathbf{m}) \right]_j = \left(\sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \left(\frac{X_{i,j} - x_j}{h_j} \right) (Y_i - m(\mathbf{X}_i)) \right), \quad 1 \leq j$$

$$\begin{aligned} \zeta_{ni}(\mathbf{x}) = & \left(\sum_{l=1}^d b_l(\mathbf{x}) K(\mathbf{X}_i - \mathbf{x}; \mathbf{h}) \left(\frac{X_{i,l} - x_l}{h_l} \right) (Y_i - m(\mathbf{X}_i)) \right) \\ & + b_0(\mathbf{x}) K(\mathbf{X}_i - \mathbf{x}; \mathbf{h})(Y_i - m(\mathbf{X}_i)) \end{aligned}$$

Asymptotic normality

$$E\zeta_{ni}(\mathbf{x}) = 0$$

$$E\zeta_{ni}^2(\mathbf{x}) = \frac{\sigma^2(\mathbf{x})p(\mathbf{x})}{h_1 \dots h_d} (b(\mathbf{x})^T \mathbf{S}(\mathbf{x}) b(\mathbf{x}) + o_p(1))$$

$$b(\mathbf{x})^T b(\mathbf{x}) = a^T \left[\mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} \right]^{-1/2} \mathbf{M}(\mathbf{x})^{-1} \mathbf{M}(\mathbf{x})^{-1} \left[\mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1} \right]^{-1/2}$$

$$a \frac{h_1 \dots h_d}{n \sigma^2(\mathbf{x}) p(\mathbf{x})} = a^T a \cdot \text{tr}(\mathbf{S}^{-1}(\mathbf{x})) \frac{h_1 \dots h_d}{n \sigma^2(\mathbf{x}) p(\mathbf{x})}$$

Therefore, for any $a \in \mathbb{R}^{d+1}$,

$$s_n^2(\mathbf{x}) = \sum_{i=1}^n \text{var}(\zeta_{ni}(\mathbf{x})) = \frac{n}{h_1 \dots h_d} \text{tr}(\mathbf{S}(\mathbf{x})) b(\mathbf{x})^T b(\mathbf{x}) = a^T a + o_p(1)$$

Asymptotic normality

$$\begin{aligned}
 \sum_{i=1}^n E[|\zeta_{ni}(\mathbf{x})|^{2+\delta}] &= n \left(\sqrt{\frac{h_1 \dots h_d}{n}} \right)^{2+\delta} E[(Y_1 - m(\mathbf{X}_1))^{2+\delta} K(\mathbf{X}_1 - \mathbf{x}; \mathbf{h})^{2+\delta} \\
 &\quad \left(b_0(\mathbf{x}) + \sum_{l=1}^d b_l(\mathbf{x}) \left(\frac{X_{1,l} - x_l}{h_l} \right) \right)^{2+\delta}] \\
 &\sim (nh_1 \dots h_d)^{-\delta/2} (K^{2+\delta})_{\mathbf{h}}^- * \left(b_0(\mathbf{x}) + \sum b_l(\mathbf{x}) \frac{x_l - x_l}{h_l} p(\cdot) \right)^{2+\delta} \\
 &\sim (nh_1 \dots h_d)^{-\delta/2} \rightarrow 0
 \end{aligned}$$

For any $a \in \mathbb{R}^{d+1}$,

$$a^T \mathbf{Z}_n(\mathbf{x}) \xrightarrow{d} N(0, a^T a)$$

$$\mathbf{Z}_n(\mathbf{x}) \xrightarrow{d} N(0, I_{d+1})$$

$$\begin{aligned}\hat{m}(\mathbf{x}) &= \arg \min_{a \in \mathbb{R}} n^{-1} \sum_{i=1}^n (Y_i - a)^2 \prod_{\ell=1}^d K_{h_\ell}(X_{i,\ell} - x_\ell) \\ &= \frac{n^{-1} \sum_{i=1}^n Y_i \cdot \prod_{\ell=1}^d K_{h_\ell}(X_{i,\ell} - x_\ell)}{n^{-1} \sum_{i=1}^n \prod_{\ell=1}^d K_{h_\ell}(X_{i,\ell} - x_\ell)}.\end{aligned}$$

Regarding $\mathbf{M}(\mathbf{x})^{-1} \mathbf{S}(\mathbf{x}) \mathbf{M}(\mathbf{x})^{-1}$,

$$\text{var}(\hat{m}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n) = n^{-1} h_1^{-1} \dots h_d^{-1} \cdot \frac{\prod_{\ell=1}^d \kappa_0(x_\ell)}{\prod_{\ell=1}^d \mu_0(x_\ell)^2} \cdot \frac{\sigma^2(\mathbf{x})}{p(\mathbf{x})} + o_p(n^{-1} h_1^{-1} \dots h_d^{-1}).$$

Multivariate N-W

In case of symmetric K , $x_j \notin [h_j, 1 - h_j]$ for $j \in J$, $x_i \in [h_i, 1 - h_i]$ for $i \notin J$

$$\mathbb{E}(\hat{m}(\mathbf{x}) \mid \mathbb{X}_n) - m(\mathbf{x}) = \frac{1}{\hat{p}(\mathbf{x})} \frac{1}{n} \sum_{i=1}^n K(\mathbf{X}_i - \mathbf{x}; \mathbf{h})(m(\mathbf{X}_i) - m(\mathbf{x}))$$

$$\begin{aligned} \mathbb{E}[K(\mathbf{X} - \mathbf{x}; \mathbf{h})(m(\mathbf{X}) - m(\mathbf{x}))] &= \int K(\mathbf{u} - \mathbf{x}; \mathbf{h})(m(\mathbf{u}) - m(\mathbf{x}))p(\mathbf{u})d\mathbf{u} \\ &= \int_{-\frac{x_d}{h_d}}^{\frac{1-x_d}{h_d}} \dots \int_{-\frac{x_1}{h_1}}^{\frac{1-x_1}{h_1}} K(t_1) \dots K(t_d) \{D^1 m(\mathbf{x})[\mathbf{h}\mathbf{t}] + \frac{1}{2}D^2 m(\mathbf{x})[(\mathbf{h}\mathbf{t})^2] + \\ &\quad \int_0^1 (1-w)(D^2 m(\mathbf{x} + \mathbf{h}\mathbf{t}w) - D^2 m(\mathbf{x}))[(\mathbf{h}\mathbf{t})^2]dw\} p(\mathbf{x} + \mathbf{h}\mathbf{t})dt_1 \dots dt_d \\ &\leq C(\max_{j \notin J} h_j + \max_{j \in J} h_j^2) \end{aligned}$$

Multivariate N-W

Using $\sup_{\mathbf{x} \in [\mathbf{h}, \mathbf{1}-\mathbf{h}]} \mathbb{E}[|\hat{m}(\mathbf{x}) - m(\mathbf{x})|^2 | \mathbb{X}_n],$

$$\int_{[\mathbf{h}, \mathbf{1}-\mathbf{h}]} \|\hat{m}(\mathbf{x}) - m(\mathbf{x})\|^2 d\mathbf{x} = O_p \left(\max_j h_j^4 + n^{-1} h_1^{-1} \cdots h_d^{-1} \right).$$

Using $\int_{[0,1]^d / [\mathbf{h}, \mathbf{1}-\mathbf{h}]} \|\hat{m}(\mathbf{x}) - m(\mathbf{x})\|^2 d\mathbf{x} \leq d(\max h_j) O_p(\max h_j^2 + \frac{1}{nh_1 \cdots h_d}),$

$$\int_{[0,1]^d} \|\hat{m}(\mathbf{x}) - m(\mathbf{x})\|^2 d\mathbf{x} = O_p \left(\max_j h_j^4 + n^{-1} h_1^{-1} \cdots h_d^{-1} \right),$$

Multivariate N-W

Using $\sup |\mathbb{E}(\hat{m}(\mathbf{x})|\mathbb{X}_n) - \hat{m}(\mathbf{x})| = O_p(\sqrt{\frac{\log n}{nh_1 \dots h_d}})$,

$$\sup_{\mathbf{x} \in [0,1]^d} |\hat{m}(\mathbf{x}) - m(\mathbf{x})| = O_p \left(\max_j h_j + n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \sqrt{\log n} \right),$$

$$\sup_{\mathbf{x} \in [\mathbf{h}, \mathbf{1}-\mathbf{h}]} |\hat{m}(\mathbf{x}) - m(\mathbf{x})| = O_p \left(\max_j h_j^2 + n^{-1/2} h_1^{-1/2} \dots h_d^{-1/2} \sqrt{\log n} \right).$$

Curse of dimensionality

Suppose that $h_j \sim n^{-1/(d+4)}$, which is the optimal size of h_j for the local linear estimator of m (and also for the N-W in the interior). Then, with this size for h_j , we have

$$\text{MSE}(\hat{m}(\mathbf{x}) \mid \mathbf{X}_1, \dots, \mathbf{X}_n) = O_p(n^{-4/(d+4)})$$

for all $\mathbf{x} \in [0, 1]^d$ in case $\hat{m}$ is the local linear estimator (for all interior $\mathbf{x} \in [\mathbf{h}, \mathbf{1} - \mathbf{h}]$ in case $\hat{m}$ is the N-W estimator).

The rate of convergence deteriorates as d increases. This phenomenon is termed as the *curse of dimensionality*.